هيئة تقويم التعليم والتدريب
Education & Training Evaluation Commission

T-104
2022

Course Specification

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Course Specification

Course Title:	Optimization Algorithms
Course Code:	TBD
Program:	Computer Science / Artificial Intelligence
Department:	Computer Science
College:	College of Computer Science
Institution:	Najran University
Version:	1
Last Revision Date:	

Table of Contents:

A. General information about the course:

Course Identification					
1. Credit hours:	3 hours				
2. Course type					
a.	University ☐	College ☐	Department☐	Track☒	Others☐
b.	Required ☒	Elective☐			
3. Level/year at which this course is offered: 3rd Year, Term2					
4. Course general Description This highly rigorous course establishes the mathematical and algorithmic foundations of optimization, the core engine of modern Artificial Intelligence and Machine Learning. Students will rigorously analyze and implement both deterministic and stochastic optimization techniques. Topics include convexity, first- and second-order unconstrained optimization (Gradient Descent, Newton's method, BFGS), constrained optimization (KKT conditions, Augmented Lagrangians), and the stochastic methods driving deep learning (SGD, Adam, RMSProp). Advanced modules cover derivative-free and Bayesian Optimization for hyperparameter tuning.					
5. Pre-requirements for this course (if any): Linear Algebra, Calculus, Introduction to Machine Learning					
6. Co- requirements for this course (if any): After completing this course, the student will be able to: • Formulate complex AI challenges as formal mathematical optimization problems. • Mathematically prove convergence properties of first and second-order algorithms. • Implement scalable optimization algorithms from scratch to train neural networks. • Diagnose and resolve optimization bottlenecks (e.g., saddle points, vanishing gradients, ill-conditioning) in deep learning architectures. • Apply Bayesian and derivative-free optimization for automated hyperparameter tuning.					

1. Teaching mode (mark all that apply)

No	Mode of Instruction	Contact Hours	Percentage
1.	Traditional classroom	45	100
2.			
3.			
4.			

2. Contact Hours (based on the academic semester)

No	Activity	Contact Hours
1.	Lectures	30
2.	Laboratory/Studio	15
3.	**Field**	
4.	Tutorial	
5.	Others (specify)	
	Total	**45**

B. Course Learning Outcomes (CLOs), Teaching Strategies and Assessment Methods

Code	Course Learning Outcomes	Code of CLOs aligned with program	Teaching Strategies	Assessment Methods
1.0	**Knowledge and understanding**			
1.1	Articulate the mathematical foundations of convex sets, functions, gradients, and Hessians.	K1	Lectures, Class Discussion	MidTermExam, Final Exam,Quizes
1.2	Formulate unconstrained and constrained optimization problems, including KKT conditions and duality.	K2	Lectures, Problem Solving	Final Exam,Quizes
..				
2.0	Skills			
2.1	Implement first-order, second-order, and quasi-Newton (BFGS) algorithms from scratch.	S1	Lab Activities, Code Reviews	Lab Assignments, Project
2.2	Construct and tune stochastic optimization routines (SGD, Adam) for training Deep Neural Networks	S2	Lab Activities, Live Coding	Lab Assignments, Project
2.3	Design Bayesian Optimization loops for hyperparameter tuning of black-box AI models.	S3	Guided Lab Projects	Final Applied Project
2.4	Mathematically analyze algorithmic complexity and theoretical convergence rates.	S4	Analytical Problem Solving	Midterm Exam, Final Exam
3.0	Values, autonomy, and responsibility			
3.1	Demonstrate academic integrity and rigorous scientific methodology in algorithm evaluation.	V1	Group Project	roject Report Evaluation
3.2	Think critically to select the optimal algorithm balancing computational cost and accuracy for specific AI tasks.	V2	Case Study Problems	Exam, Case Study

C. Course Content

No	List of Topics	Contact Hours
1.	Mathematical Foundations: Vector calculus, Jacobians, Hessians, and Taylor Expansions	3
2.	Convex Optimization Basics: Convex sets, convex functions, and global minima	3
3	Unconstrained Optimization I: Gradient Descent, Line Search (Armijo conditions), Convergence Analysis	3
4	Unconstrained Optimization II: Newton's Method, Secant Methods, and limitations	3
5	Unconstrained Optimization III: Quasi-Newton Methods (BFGS, L-BFGS) and Conjugate Gradient	3
6	Automatic Differentiation: Forward and reverse mode (Backpropagation in AI)	3
7	Stochastic Optimization I: Stochastic Gradient Descent (SGD), Mini-batching, and Variance Reduction	3
8	Stochastic Optimization II: Adaptive learning rates (AdaGrad, RMSProp, Adam, AdamW)	3
9	Optimization in Deep Learning: Escaping saddle points, ill-conditioning, and landscape analysis	3
10	Constrained Optimization I: Lagrange multipliers and the Karush-Kuhn-Tucker (KKT) conditions	3
11	Constrained Optimization II: Penalty methods, Barrier methods, and Augmented Lagrangians	3
12	Duality and AI Applications: Dual formulation and Support Vector Machines (SVMs)	3
13	Derivative-Free Optimization: Nelder-Mead and Black-Box scenarios	3
14	Bayesian Optimization: Gaussian Processes and Acquisition Functions (UCB, Expected Improvement)	3
15	Metaheuristics in AI: Evolutionary Strategies and Particle Swarm Optimization	3
Total		45

D. Students Assessment Activities

No	Assessment Activities *	Assessment timing (in week no)	Percentage of Total Assessment Score
1.	Mid-Term Exam	Week 7	20%
2.	Advanced Lab Assignments (Algorithm Implementation)	Weeks 3, 5, 9, 11	20%
3.	Quizzes	TBD	10%
4.	Applied Term Project (Optimizing a Deep Learning Architecture)	Week 14	10%
4.	Final Exam	Week 16	40%

*Assessment Activities (i.e., Written test, oral test, oral presentation, group project, essay, etc.)

E. Learning Resources and Facilities

1. References and Learning Resources

Essential References	*Convex Optimization* by Stephen Boyd and Lieven Vandenberghe (Cambridge University Press).
Supportive References	*Optimization for Machine Learning* by Suvrit Sra, Sebastian Nowozin, and Stephen J. Wright (MIT Press). *Deep Learning* (Chapter 8: Optimization for Training Deep Models) by Ian Goodfellow, Yoshua Bengio, and Aaron Courville.
Electronic Materials	University LMS, Lecture slides adapted from standard global frameworks (e.g., TU Berlin, Stanford CS229/CS330).
Other Learning Materials	Jupyter Notebooks / Google Colab for hands-on algorithm implementation in Python (using NumPy and PyTorch).

2. Required Facilities and equipment

Items	Resources
facilities **(Classrooms, laboratories, exhibition rooms, simulation rooms, etc.)**	• Lecture room with 25 to 30 student accommodation. • Computer Lab with all essentials to accommodate 20 students.
Technology equipment **(projector, smart board, software)**	• Each lecture room is equipped with the latest computing devices and projectors. • Information technology laboratory rooms equipped with high-performance computers for matrix computations. • Faculty member provided with necessary computational resources.
Other equipment **(depending on the nature of the specialty)**	Access to cloud GPUs (e.g., AWS, GCP, or local university clusters) for testing stochastic optimization algorithms on large datasets.

F. Assessment of Course Quality

Assessment Areas/Issues	Assessor	Assessment Methods
Effectiveness of teaching	Students, External reviewers` visit from Accreditation Agency	Survey Formal Classroom Observation
Effectiveness of students assessment	Quality and Development Unit, Curriculum Committee,	Teachers` feedback, Students feedback, Course report
Quality of learning resources	Quality and Development Unit	Annual quality improvement program review
The extent to which CLOs have been achieved	Quality and Development Unit	Course report, data analysis of achievement test
Other		

Assessor (Students, Faculty, Program Leaders, Peer Reviewer, Others (specify)

Assessment Methods (Direct, Indirect)

G. Specification Approval Data

COUNCIL /COMMITTEE	DEPARTMENT COUNCIL OF COMPUTER SCIENCE
REFERENCE NO.	
DATE	